# AI-POWERED TRAINING & PLACEMENT MANAGEMENT SYSTEM (AI-TPMS)

**Complete Technical Architecture & Concept Document | Final Year Academic Project**

## 1. Executive Summary

AI-TPMS is an enterprise-grade academic placement and career development platform that automates the campus placement lifecycle. Engineered with FastAPI, SQLAlchemy 2.0, Modern HTML5/Tailwind/Lucide, and Multi-Modal AI engines, the system replaces manual T&P; workflows with automated candidate screening, sequential aptitude testing, technical evaluations, and real-time proctored mock interviews.

## 2. Explainable Multi-Modal AI Subsystems

| Subsystem | Underlying Technology | Key Deliverables / Capabilities |
|---|---|---|
| Resume Intelligence | PDF Tokenizer & Symbol Parser | 98%+ precision on C++, C#, SQL; Google X-Y-Z scoring. |
| Cognitive Aptitude | Dynamic Question Generator | 20-minute timed adaptive assessment with 65% cutoff gating. |
| Technical Core MCQ | Taxonomy Knowledge Graph | DSA, DBMS (ACID, Indexes), OS, OOPs, Computer Networks. |
| Computer Vision Proctor | YOLOv8 & OpenCV | Detects smartphones, unauthorized books, and multiple persons. |
| Head Pose & Eye Gaze | MediaPipe 3D Landmark Mesh | Calculates Pitch, Yaw, Roll angles and out-of-screen deflections. |
| Speech & Fluency | RIFF WAV & Silence Detector | Analyzes Words-Per-Minute (WPM), hesitation pauses, filler words. |
| Mitra AI Mentor | Dual-Engine (Gemini + KB) | Conversational guidance for rounds, eligibility, and 47 job drives. |

## 3. 3-Round Sequential Placement Pipeline

Candidate progression is governed by deterministic threshold gating. A student must achieve the minimum qualifying score in each phase to unlock subsequent evaluations:

- **Round 1 (Aptitude & Logical Mastery):** 20 MCQs covering Quantitative, Logical, and Verbal. Score ≥ 65% required.
- **Round 2 (Technical Core Assessment):** 20 MCQs evaluating Data Structures, DBMS, OS, and Networks. Score ≥ 70% required.
- **Round 3 (AI Mock Interview Room):** Interactive video/audio room evaluating STAR responses with computer vision and audio telemetry.

## 4. 47 Corporate Placement Drives & Salary Matrix

AI-TPMS features 47 pre-configured corporate recruitment drives across Tier-1 Super Dream (Google, Microsoft, Amazon, Stripe - up to 28.5 LPA), Tier-1 FinTech (Goldman Sachs, Morgan Stanley - 18 to 24 LPA), High-Growth Cloud (Oracle, Cisco, Salesforce - 15 to 20 LPA), and Global Consulting (Deloitte, Accenture, TCS, Infosys - 6.5 to 12 LPA).